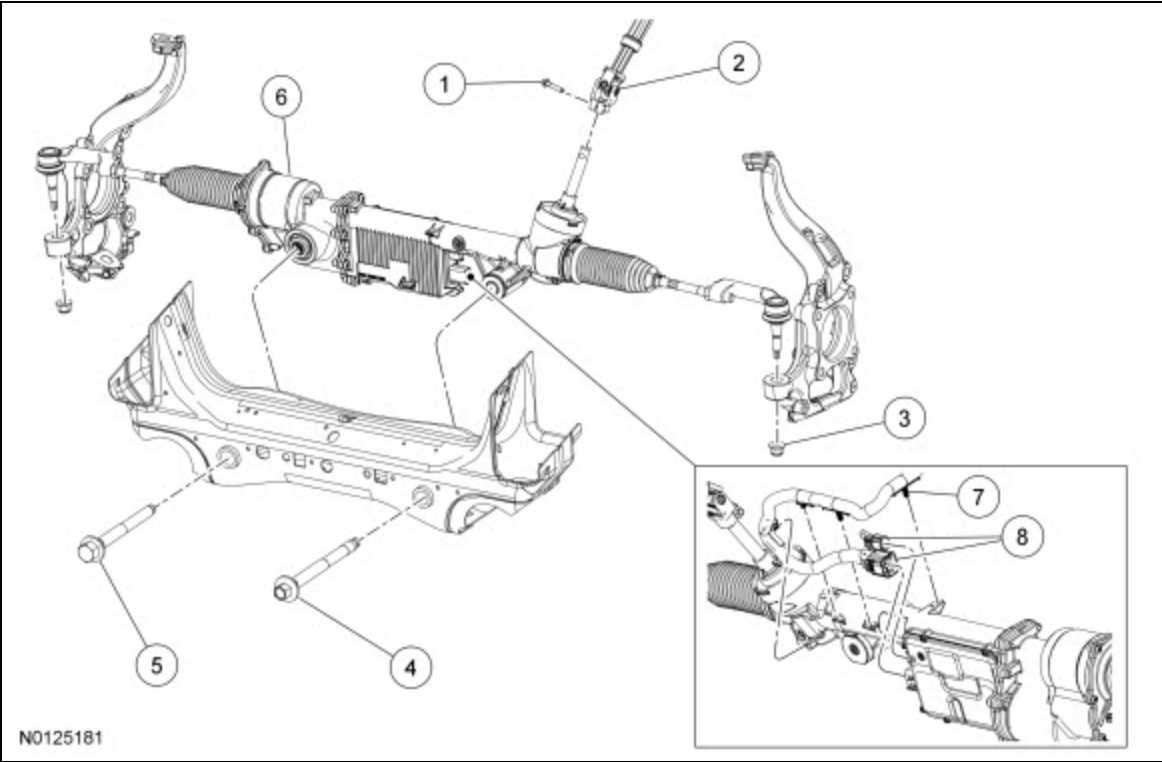


Steering Gear — Electronic Power Assist Steering (EPAS)

 [Ford Ecat Electronic Parts Catalog](#)

Special Tool(s)

 ST2945-A	Separator, Ball Joint Tool 204-592
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Item	Part Number	Description
1	N808684 N808684	Steering column shaft-to-steering gear bolt
2	3C662 3C662	Steering column shaft
3	W520215 W520215	Outer tie-rod end nut (2 required)
4	W713954 W713954	Steering gear-to-crossmember bolt
5	W715436 W715436	Steering gear-to-crossmember bolt
6	3504 3504	Steering gear
7	—	Steering gear wiring harness retainers (4 required) (part of 3F720)
8	—	Steering gear electrical connectors (2 required) (part of 3F720)

1. If installing a new steering gear, connect the scan tool and upload the module configuration information from the Power Steering Control Module (PSCM) . For additional information, refer to Section 418-01.

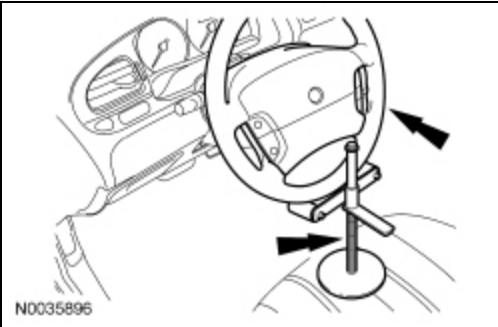
2. **NOTICE: Disconnect the negative battery cable anytime the steering gear is being serviced or damage to the steering gear internal power relay may occur resulting in steering gear replacement.**

Disconnect the battery ground cable. For additional information, refer to Section 414-01.

3. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to Section 100-02.

4. **NOTICE: Do not allow the steering column shaft to rotate while disconnected from the gear or damage to the clockspring may occur. If there is evidence that the steering column shaft has rotated, remove and recenter the clockspring. For additional information, refer to Section 501-20B.**

Use a suitable holding device to hold the steering wheel in the straight-ahead position.



5. Remove the front wheels and tires. For additional information, refer to Section 204-04.

6. Disconnect the 2 steering gear electrical connectors, the 4 wiring harness retainers and position the wiring harness aside.

7. If equipped, remove the skid plate bolts and the skid plate.

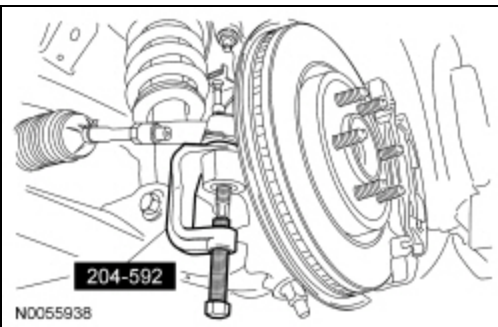
- To install, tighten to 25 Nm (18 lb-ft).

8. Remove and discard the outer tie-rod end nuts.

- To install, tighten to 115 Nm (85 lb-ft).

9. **NOTICE: Use care when installing the Ball Joint Tool Separator or damage to the tie-rod end boot may occur.**

Using the Ball Joint Tool Separator, disconnect the tie-rod ends from the wheel knuckles.



10. **NOTICE: Do not allow the steering column shaft to rotate while disconnected from the gear or damage to the clockspring may occur. If there is evidence that the steering column shaft has rotated, remove and recenter the clockspring. For additional information, refer to Section 501-20B.**

Remove and discard the steering column shaft-to-steering gear bolt and disconnect the steering column shaft from the steering gear.

- To install, tighten to 30 Nm (22 lb-ft).

11. Remove and discard the 2 steering gear-to-crossmember bolts and remove the steering gear.

- To install, tighten to 440 Nm (325 lb-ft).

12. NOTICE: Make sure the tie-rod end boots are seated correctly on the tie-rod ends or tie-rod end failure may occur.

To install, reverse the removal procedure.

13. When installing a new steering gear, it must be configured (using vehicle as-built data or module configuration information retrieved earlier in this procedure). For additional information on configuration, refer to Programmable Module Installation (PMI) in Section 418-01.

14. Check and, if necessary, adjust the front toe. For additional information, refer to Section 204-00.